

EDUCATION

MILA QUEBEC | ETS MONTREAL

Research MSc in Computer Science (GPA: 4.15/4.30)

Montreal, CA
Aug 2021 - Present

- **Thesis:** Sample efficient reinforcement learning for real world systems. Supervised by Samira E. Kahou
- **Awards:** NeurIPS Conference Grant (worth 500USD). ICML Participation Grant (worth 500USD). Tuition scholarship worth 16000 CAD given to top 10% of program. Amii Talent Bursary for Emerging Researchers worth 1500CAD.

NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI (NITT)

B.Tech in Instrumentation and Control Engineering (GPA: 9.21/10, Rank: 6/103)

Tiruchirappalli, India
Aug 2017 - May 2021

- **Thesis:** Data Driven Fuzzy Control Using Distending Functions

PROJECTS

IMPROVING SAMPLE EFFICIENCY THROUGH THE REDUCIBLE LOSS: MILA, QUEBEC

Supervisor: Samira E. Kahou

Montreal
May. 2022 - Present

- Devised the Reducible Loss (ReLo) prioritization that improves performance by 20% over vanilla prioritized experience replay, increasing the sample efficiency and accuracy of the value function. Accepted to the **DeepRL Workshop, NeurIPS 2022**.
- ReLo can be integrated in **any** off policy algorithm with minimal code changes and negligible computational overhead.

MODEL SELECTION METHODS FOR OFFLINE RL: MILA, QUEBEC

Supervisor: Samira E. Kahou, Jörg Bornschein (DeepMind)

Montreal
Feb. 2022 - Present

- Proposed a metric inspired by prequential MDL (pMDL) to bridge the gap in evaluation strategies between online and offline RL. Accepted at the **Offline RL Workshop, NeurIPS 2022**.
- pMDL addresses the absence of standard methods in offline RL for comparing algorithms simultaneously on their sample efficiency and speed of adaptation to distribution shifts

REPRESENTATION LEARNING FOR MODEL BASED RL: MILA, QUEBEC

Supervisor: Samira E. Kahou, Danijar Hafner (Google Brain)

Montreal
Jan. 2021 - Feb 2022

- Improved sample efficiency of training by **2x fewer** gradient steps through Variational Sparse Gating (VSG). Accepted at **NeurIPS 2022**.
- Built the BringBackShapes environment that tests agents on long horizon tasks with partial observability and stochasticity.

RL BASED ADAPTIVE CONTROLLER TUNING FOR NON LINEAR SYSTEMS: NITT

Undergraduate Research Assistant, Supervisor: K Srinivasan

Trichy, India
Sept. 2020 - May. 2021

- Designed the mPPO adaptive PID controller that achieves **15%** lower tracking error compared to hand crafted controller. Accepted at **Applied Soft Computing 2022**.
- Trained agent is robust to process parameter deviations upto **10%** while maintaining competitive performance.

EXPERIENCE

ENVIROCAR PROJECT, 52°NORTH - GOOGLE SUMMER OF CODE

Software Development Intern

Münster, Germany
May 2019 - Sept. 2019

- Developed an analysis tool for the Android client that detailed driving patterns of users, such as distance covered, emissions, fuel consumption, over given time periods obtained from vehicular data.
- Designed a visualisation to allow users to identify hotspots in emissions & fuel consumption using the MapBox SDK.

PUBLICATIONS

Learning Robust Dynamics through Variational Sparse Gating: Paper  Code 

NeurIPS 2022

Arnav Jain, Shivakanth Sujit, Shruti Joshi, Vincent Michalski, Danijar Hafner, Samira E Kahou.

Bridging the Gap Between Offline and Online Reinforcement Learning Evaluation Methodologies: Paper  Code 

Offline RL Workshop, NeurIPS 2022

Shivakanth Sujit, Pedro Braga, Jörg Bornschein, Samira E Kahou

Prioritizing Samples in Reinforcement Learning with Reducible Loss: Paper  Code 

DeepRL Workshop, NeurIPS 2022

Shivakanth Sujit, Somjit Nath, Pedro Braga, Samira E Kahou

Reinforcement learning based model free adaptive controller for linear and nonlinear unstable processes: Paper  Code 

Applied Soft Computing 2021

Shuprajhaa Thayumanavan, Shivakanth Sujit & Srinivasan K.